Sie, Deta Dirganjaya

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Research Interest

Networked systems, network telemetry, ML-for-network, network-for-ML, distributed systems, operating systems, storage systems.

Education

Dian Nuswantoro University Bachelor of Computer Science
#5 Best Innovation University in Indonesia[1] *Sept. 2022 – Jun 2026*

- Cumulative GPA: 3.93 out of 4.00
- Major GPA: 3.92 out of 4.00

Research Experience

International Research Collaboration May. 2025 – Present
University of Maryland, College Park - FROOT-Lab *Remote*

- Collaborated with **Alan Zaoxing Liu** and his Ph.D. students to conduct research on Approximation-First Cloud Telemetry.
- Designed and implemented a high-performance sketch-based telemetry system for approximate data analytics.
- Developed a full ingestion–storage–query pipeline with windowed aggregation, WAL buffering, and a custom PromQL-compatible query engine for real-time analysis.
- Explored and implemented advanced optimizations such as deterministic sketch merging, direct compression, and tolerance-aware query planning to enhance scalability and cost efficiency.

University of Chicago - Indonesia Research Training Program Jan. 2025 – Jun. 2025

- Selected as one of the top 50 Indonesian students in computer science and related fields for an international research training program led by **Haryadi S. Gunawi** of **University of Chicago.**
- Covered 30+ technical papers from SIGCOMM and NSDI conferences.
- Replicated and reproduced key experiments from top-tier publications.
  - Reproduced and enhanced **OctoSketch: Enabling Real-Time, Continuous Network Monitoring over Multiple Cores** on Chameleon Cloud with Exponential Moving Average (EMA), achieving up to 0.89 accuracy and more stable per-flow estimation.
  - Reproduced and extended **ModelKeeper: Accelerating DNN Training via Automated Training Warmup** with packet-level load balancing inspired by MLT, using MTU-based weight chunking and ECMP transfer to reduce warmup bottlenecks in distributed model loading.
- Received intensive training in research methodologies, technical writing, and engagement with top-tier research communities.

Research Assistant Aug. 2024 – Jan. 2025
Dian Nuswantoro University Research Center, Dian Nuswantoro University *Hybrid, Semarang*

- Research assistant under **Guruh Fajar Shidik** on Artificial Intelligence (AI), Distributed Systems, Image Processing, and Distributed Computing.
- Optimizing workload distribution based on CPU and RAM utilization using the Cluster RL Q-Learning technique in a Federated Edge Cloud (FEC) environment simulated with CloudSimSDN.
- Analyzing and testing the impact of the "selective" parameter frequency in the SSVSG method on Named Entity Recognition (NER) for Indonesian disaster dataset.

Research Assistant Aug. 2024 – Jan. 2025
AI Research and Development Group, Dian Nuswantoro University *On-site, Semarang*

- Research group under the Faculty of Computer Science, Dian Nuswantoro University, led by **Adhitya Nugraha**, focusing on Artificial Intelligence (AI).
- Developed a Virtual Smart Assistant (VSA) system integrating real-time facial recognition and emotion detection with an LLM-based chatbot, enabling users to interact seamlessly through text-to-speech technology.
- Collaborated with the Faculty of Health on research to prevent cyberbullying on Indonesian social media using a Multi-Modal machine learning framework.

Work Experience

Teaching Assistant Aug. 2024 – Jan. 2025

AI Research and Development Group *On-Site, Semarang*

- Assisting and teaching students in a career guidance class, preparing them to enter the workforce.
- Responsible for teaching Data Science classes with a total of 50+ students.
- Provided subject matter on data collection, data pre-processing, data visualization, and modeling.

Business Analyst Feb. 2023 – Jul. 2023

CV Sanjaya Utama *Remote, Semarang*

- Developed a system for generating product bundle promotions used by the company for product marketing using the association rule mining.
- Responsible for data analysis, modeling, visualization, maintenance, and development of models to support business decision-making.

Projects

VSA: Virtual Smart Assistant

- A personal assistant designed to support and assist students of Dian Nuswantoro University
- Utilizing facial recognition and emotion tracking to detect students' emotional states.
- Powered by LLM-based conversational AI to provide appropriate responses based on students' emotions.
- Integrated with text-to-speech technology for a more natural and engaging communication experience.
- Contributed 1000+ LOC (3200+ lines of group code).
- Tools used: Python, JavaScript, Vite, ONNX, CUDA, Jetson Nano, DGX A100.

Optimizing ANN with Grid Search for Predicting Stunting Among Toddlers

- Investigates the application of Artificial Neural Networks (ANN) in predicting stunting, highlighting the impact of hyperparameter normalization and optimization on model accuracy and reliability.
- Utilizes SMOTE for dataset balancing and MinMaxScaler for normalization to enhance model performance.
- GridSearchCV optimization enables the ANN model to achieve an improvement 6.43% in accuracy, 8.64% in recall, and a 6.41% in F1-score, surpassing other comparative models.
- Contributed 600+ LOC.
- Tools used: Python.

CLCM: Custom Lightweight CNN Model for low-end edge devices

- CNN-based face recognition model optimized for high efficiency and real-time performance on low-power devices.
- Outperforms well-known pre-trained models, delivering three times faster inference speed, with an average real-time detection time of 0.013 seconds.
- Efficiently utilizes up to 90% of the Jetson Nano's GPU, ensuring optimal processing power for edge computing.
- Contributed 1000+ LOC.
- Tools used: Python, ONNX, CUDA, Jetson Nano.

Prior Publications

1. **Sie, Deta Dirganjaya**, Guruh Fajar Shidik, Maulidya Ayu Ardiena, Radhitya Marendratama, and Edi Jaya Kusuma **"Analyzing the Impact of Transpose Layers on CNN-Based Deep Learning"**, *iSemantic 2025, Accepted for publication.*

Awards

Flagship Program Student of Department of Computer Science Year 2023 Aug. 2023

- Selected as one of the top 50 out of 1855 computer science students in Dian Nuswantoro University Batch 2022 Flagship Program, an acceleration program for high-achieving students.

Certificates

NVIDIA: Deep Learning Training

- Learned deep learning fundamentals including neural networks, CNNs, and RNNs.
- Enhance model accuracy with data augmentation, transfer learning, and pre-trained models.
- Deploy deep learning models efficiently using NVIDIA tools and frameworks.
- 10+ hours of hands-on, self-paced training with real-world applications.

CCNA: Enterprise Networking, Security, and Automation

- Designed and simulated enterprise networks with BGP for multi-domain routing.
- Configured advanced NAT and WAN security, including VPNs, ACLs, and firewalls.
- Automated network operations using Python, REST APIs, and Cisco DNA Center.
- Explored SDN and virtualization for modern network management.
- 140+ hours of lab practice.

CCNA: Switching, Routing, and Wireless Essentials

- Designed and simulated enterprise networks with VLANs, trunking, and inter-VLAN routing.
- Implemented redundancy solutions using STP variants and EtherChannel for reliability.
- Configured and secured wireless networks, including WLAN architectures and troubleshooting.
- Applied security best practices for device hardening and network segmentation.
- 90+ hours of lab practice.

CCNA: Introduction to Networks

- Learn networking fundamentals, including devices, media, and protocols.
- Build and configure LANs with essential network devices.
- Understand IP addressing and network services like DHCP and DNS.
- 50+ hours of lab practice.

Technical Skills

Programming Languages: C, C++, Go, Rust, Python, Java, SQL.

Databases: RocksDB, InfluxDB, MongoDB, MySQL, SQLite.

Libraries: TensorFlow, PyTorch, TensorRT, ONNX, NumPy, Ray, Net, OpenCV, SciPy, MLxtend, Seaborn, Matplotlib, Pandas.

Tools: Linux, WSL, Kubernetes, Docker, CUDA, Prometheus, Grafana, Kafka, VictoriaMetrics, Cisco Packet Tracer, Git, pprof, Intel VTune.

Misc: NVIDIA Jetson, Raspberry Pi, NVIDIA DGX A100.

References

Alan Zaoxing Liu — *Assistant Professor*
Computer Science, University of Maryland, College Park — *zaoxing@umd.edu*

Haryadi S. Gunawi — *Professor*
Computer Science, University of Chicago — *haryadi@cs.uchicago.edu*

Guruh Fajar Shidik — *Professor & Center Director*
Computer Science, Dian Nuswantoro University Research Center — *guruh.fajar@research.dinus.ac.id*